

Advanced (Habanero)

- Q1.** Solve the system:  $x^2 + y^2 = 4$ ,  $x^2 - y^2 = -1$   
(A)  $(-1, -\sqrt{3})$   
(B)  $(-1, \sqrt{3})$   
(C)  $(1, -\sqrt{3})$   
(D)  $(1, \sqrt{3})$   
(E)  $(0, 0)$
- Q2.** The system  $y = x^2 - 2$ ,  $y = -x^2 + 4$  has how many solutions?  
(A) 0  
(B) 1  
(C) 2  
(D) 3  
(E) 4
- Q3.** What is the  $x$ -coordinate of the solution to the system  $x^2 + y^2 = 1$ ,  $y = x^2 - 1$ ?  
(A)  $-1$   
(B)  $0$   
(C)  $1$   
(D)  $-\sqrt{2}$   
(E)  $\sqrt{2}$
- Q4.** The graphs of  $y = x^2 - 1$  and  $y = -x^2 + 3$  intersect at what  $y$ -value?  
(A)  $-2$   
(B)  $-1$   
(C)  $1$   
(D)  $2$   
(E)  $0$
- Q5.** The system  $x^2 - y^2 = 1$ ,  $x^2 + y^2 = 5$  has solutions with  $x$ -coordinates  
(A)  $-2, 2$   
(B)  $-1, 1$   
(C)  $-\sqrt{2}, \sqrt{2}$   
(D)  $-\sqrt{3}, \sqrt{3}$   
(E)  $-\sqrt{6}, \sqrt{6}$
- Q6.** Solve the system:  $y = x^2 - 3$ ,  $y = -x^2 + 1$   
(A)  $(-1, -2)$   
(B)  $(-1, 2)$   
(C)  $(0, -1)$   
(D)  $(1, -2)$   
(E)  $(1, 2)$
- Q7.** The system  $y = x^2 - 2y$ ,  $y = -x^2 - 2$  has how many real solutions?  
(A) 0  
(B) 1  
(C) 2  
(D) 3  
(E) 4
- Q8.** The system  $x^2 + y^2 = 4$ ,  $x^2 - y^2 = -1$  represents which geometric figures?  
(A) Two parallel lines  
(B) Two intersecting lines  
(C) A circle and an ellipse  
(D) A circle and a hyperbola  
(E) Two circles

Open Ended Questions (FRQ)

- Q9.** Solve the system:  $x^2 + y^2 = 10$ ,  $x^2 - y^2 = 2$  and find the distance between the two solutions.
- Q10.** Graph the system  $y = x^2 + 1$ ,  $y = -x^2 - 3$  and find the  $x$ -coordinates of the points of intersection.

Chef's Tips

- Emphasize the importance of checking the validity of solutions.
- Remind students to consider multiple approaches to solving systems of quadratic equations.
- Encourage the use of graphing calculators or software to visualize the geometric representations of the systems.